

JOBSHEET PRAKTIKUM

Mata Kuliah: Pemrograman Web Lanjut

Pertemuan 15 – Implementasi Many-to-Many Relationship pada Filament

Capaian Pembelajaran

Setelah mengikuti praktikum ini, mahasiswa mampu:

1. Memahami konsep **Many-to-Many Relationship** pada database.
2. Membuat tabel relasi (pivot table) pada Laravel.
3. Menghubungkan model menggunakan belongsToMany.
4. Menggunakan **multiple select relationship** pada form Filament.
5. Membuat **Tag Resource** pada Filament Admin Panel.
6. Mengelola relasi menggunakan **Relationship Manager**.

Framework yang digunakan:

- **Laravel**
 - **Filament**
-

A. Latar Belakang

Pada sistem blog, satu **Post** dapat memiliki banyak **Tag**, dan satu **Tag** dapat digunakan oleh banyak **Post**.

Contoh:

Post	Tags
Laravel CRUD	Laravel, PHP
Laravel Tutorial	Laravel
PHP Basic	PHP

Relasi tersebut disebut **Many-to-Many Relationship**.

Diagram relasi:

Post
⇕
Many To Many
⇕
Tag

Relasi ini membutuhkan **pivot table** untuk menghubungkan kedua tabel.

✦ B. Permasalahan Penyimpanan Tag

Sebelumnya tag disimpan dalam format **JSON** di tabel posts.

Contoh:

tags

["Laravel 12","PHP"]

Kelemahan metode ini:

- Data sulit dimodifikasi
- Duplikasi data
- Tidak efisien untuk query database
- Tidak terstruktur

Solusi:

Menggunakan **Many-to-Many Relationship** dengan tabel **pivot**.

✦ C. Struktur Database Many-to-Many

1 Tabel Posts

posts

id

title

slug

category_id

image

body

published

published_at

2 Tabel Tags

tags

id

name

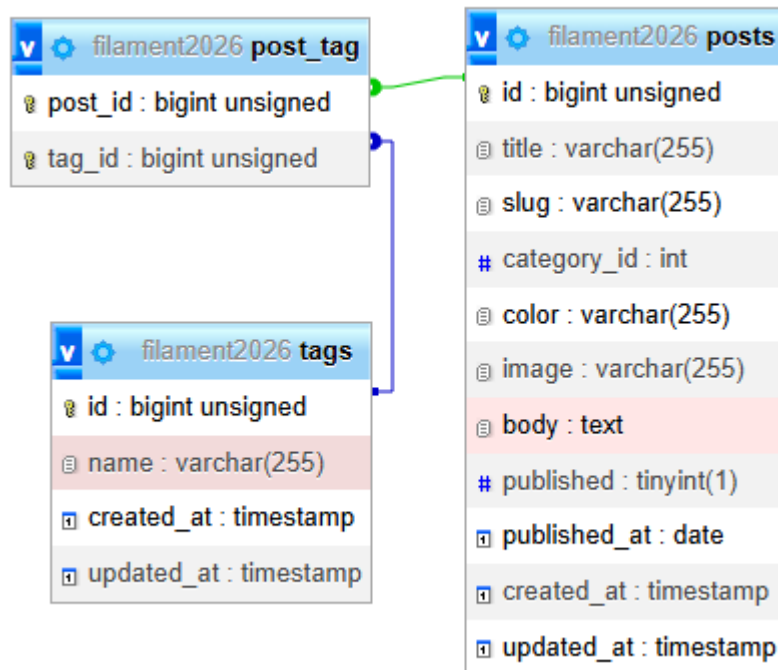
3 Pivot Table

post_tag

post_id

tag_id

Pivot table berfungsi untuk menghubungkan **posts** dan **tags**.



🔴 D. Rollback Migration

Jika sebelumnya menggunakan JSON column pada tags, rollback migration terlebih dahulu.

- `php artisan migrate:rollback`

```
PS C:\laragon\www\PraktikumPwL> php artisan migrate:rollback
```

```
INFO Rolling back migrations.
```

```
2026_03_01_143404_create_products_table ..... 15.68ms DONE
```

```
PS C:\laragon\www\PraktikumPwL> php artisan migrate:rollback
```

```
INFO Rolling back migrations.
```

```
2026_02_28_075417_create_posts_table ..... 16.23ms DONE
```

```
PS C:\laragon\www\PraktikumPwL>
```

note : kita akan melakukan rollback pada tabel prodict dan post, Untuk menentukan berapa kali rollback, kita lihat urutan timestamp migration karena Laravel rollback berdasarkan batch terakhir yang dijalankan. Sebagai contoh dari folder migrations

- 1 0001_01_01_000000_create_users_table.php
- 2 0001_01_01_000001_create_cache_table.php
- 3 0001_01_01_000002_create_jobs_table.php
- 4 2026_02_28_074211_create_categories_table.php
- 5 2026_02_28_075417_create_posts_table.php
- 6 2026_03_01_143404_create_products_table.php

Urutan eksekusi Laravel adalah dari atas ke bawah.

Jika ingin rollback posts dan products

Yang terakhir dibuat:

- **products**
- **posts**

Maka harus rollback 2 langkah.

Kemudian hapus kolom:

- `$table->json('tags');` pada `create_posts_table.php`

```
2026_02_28_075417_create_posts_table.php X
PraktikumPWL > database > migrations > 2026_02_28_075417_create_posts_table.php
1  <?php
2
3  use Illuminate\Database\Migrations\Migration;
4  use Illuminate\Database\Schema\Blueprint;
5  use Illuminate\Support\Facades\Schema;
6
7  return new class extends Migration
8  {
9      /**
10       * Run the migrations.
11       */
12     public function up(): void
13     {
14         Schema::create('posts', function (Blueprint $table) {
15             $table->id();
16             $table->string('title');
17             $table->string('slug');
18             $table->integer('category_id');
19             $table->string('color')->nullable();
20             $table->string('image')->nullable();
21             $table->text('body')->nullable();
22             $table->json('tags')->nullable();
23             $table->boolean('published')->default(false);
24             $table->date('published_at')->nullable();
25             $table->timestamps();
26         });
27     }
28
29     /**
```

E. Membuat Tabel Tags

Buka file `create_posts_table.php` dan tambahkan kode pada migration:

- ```
Schema::create('tags', function (Blueprint $table) {
 $table->id();
 $table->string('name');
 $table->timestamps();
});
```

```

PraktikumPWL > database > migrations > 2026_02_28_075417_create_posts_table.php
1 <?php
2
3 use Illuminate\Database\Migrations\Migration;
4 use Illuminate\Database\Schema\Blueprint;
5 use Illuminate\Support\Facades\Schema;
6
7 return new class extends Migration
8 {
9 /**
10 * Run the migrations.
11 */
12 public function up(): void
13 {
14 Schema::create('posts', function (Blueprint $table) {
15 $table->id();
16 $table->string('title');
17 $table->string('slug');
18 $table->integer('category_id');
19 $table->string('color')->nullable();
20 $table->string('image')->nullable();
21 $table->text('body')->nullable();
22 $table->boolean('published')->default(false);
23 $table->date('published_at')->nullable();
24 $table->timestamps();
25 });
26
27 Schema::create('tags', function (Blueprint $table) {
28 $table->id();
29 $table->string('name');
30 $table->timestamps();
31 });
32 }
33
34 /**
35 * Reverse the migrations.
36 */
37 public function down(): void
38 {
39 Schema::dropIfExists('posts');
40 }
41 };
42

```

## F. Membuat Pivot Table

Buka file create\_posts\_table.php dan tambahkan kode pada migration:

- Schema::create('post\_tag', function (Blueprint \$table) {  
     \$table->foreignId('post\_id')  
         ->constrained()  
         ->cascadeOnDelete();  
     \$table->foreignId('tag\_id')  
         ->constrained()  
         ->cascadeOnDelete();  
     \$table->primary(['post\_id', 'tag\_id']);  
 });

PraktikumPWL > database > migrations > 2026\_02\_28\_075417\_create\_posts\_table.php

```
8 {
13 {
14 Schema::create('posts', function (Blueprint $table) {
25 });
26
27 Schema::create('tags', function (Blueprint $table) {
28 $table->id();
29 $table->string('name');
30 $table->timestamps();
31 });
32
33 Schema::create('post_tag', function (Blueprint $table) {
34 $table->foreignId('post_id')
35 ->constrained()
36 ->cascadeOnDelete();
37 $table->foreignId('tag_id')
38 ->constrained()
39 ->cascadeOnDelete();
40 $table->primary(['post_id', 'tag_id']);
41 });
42 }
43
44 /**
45 * Reverse the migrations.
46 */
47 public function down(): void
48 {
49 Schema::dropIfExists('posts');
50 }
51 };
52
```

---

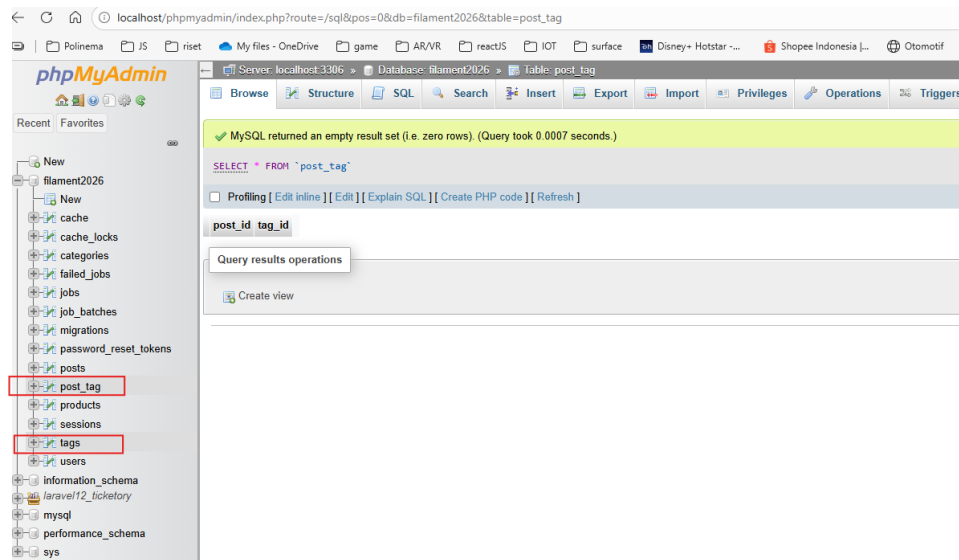
## Jalankan Migration

- php artisan migrate

• PS C:\laragon\www\PraktikumPWL> php artisan migrate

INFO Running migrations.

```
2026_02_28_075417_create_posts_table 81.46ms DONE
2026_03_01_143404_create_products_table 29.35ms DONE
```



## ✚ G. Membuat Resource Tag pada Filament

- php artisan make:filament-resource Tag
- ```
PS C:\laragon\www\PraktikumPWL> php artisan make:filament-resource Tag
```
- The "title attribute" is used to label each record in the UI.
- You can leave this blank if records do not have a title.
- What is the title attribute for this model?
- ```
> name
```
- Would you like to generate a read-only view page for the resource? (yes/no) [no]
- ```
> no
```
- Should the configuration be generated from the current database columns? (yes/no) [no]
- ```
> no
```
- Does the model use soft-deletes? (yes/no) [no]
- ```
> no
```
- INFO** Filament resource [App\Filament\Admin\Resources\Tags\TagResource] created successfully.

✚ H. Membuat Model Tag

- php artisan make:model Tag
- ```
PS C:\laragon\www\PraktikumPWL> php artisan make:model Tag
```
- INFO** Model [C:\laragon\www\PraktikumPWL\app\Models\Tag.php] created successfully.

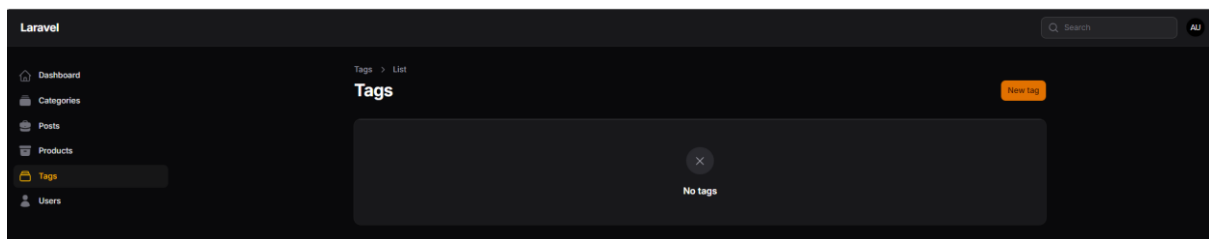
Edit models tag.php:

- protected \$fillable = [
 'name'
 ];

```

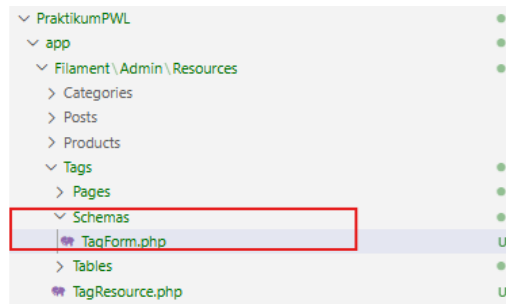
PraktikumPWL > app > Models > Tag.php
1 <?php
2
3 namespace App\Models;
4
5 use Illuminate\Database\Eloquent\Model;
6
7 Windsurf: Refactor | Explain
8 class Tag extends Model
9 {
10 protected $fillable = [
11 'name'
12];
13 }
14

```



## Form Tag

- Buka file TagForm.php pada schemas



- Tambah kode berikut
  - `TextInput::make('name')`



```

PraktikumPWL > app > Filament > Admin > Resources > Tags > Schemas > TagForm.php
1 <?php
2
3 namespace App\Filament\Admin\Resources\Tags\Schemas;
4
5 use Filament\Schemas\Schema;
6 use Filament\Forms\Components\TextInput;
7
8 class TagForm
9 {
10 Windsurf: Refactor | Explain | Generate Function Comment | X
11 public static function configure(Schema $schema): Schema
12 {
13 return $schema
14 ->components([
15 TextInput::make('name')
16]);
17 }
18 }

```

Tags > Create

## Create Tag

Name

### Table Tag

Buka file TagsTable.php pada folder Tags



Tambahkan kode berikut

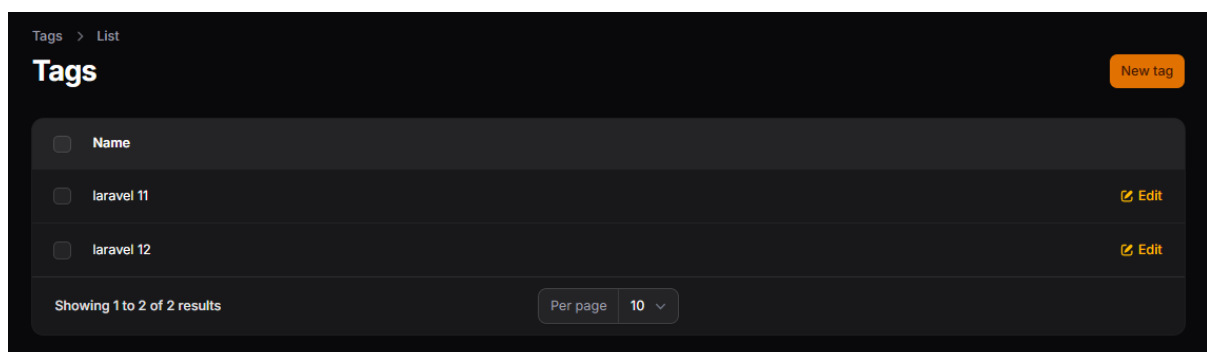
- `TextColumn::make('name')`

PraktikumPWL > app > Filament > Admin > Resources > Tags > Tables > TagsTable.php

```
1 <?php
2
3 namespace App\Filament\Admin\Resources\Tags\Tables;
4
5 use Filament\Actions\BulkActionGroup;
6 use Filament\Actions\DeleteBulkAction;
7 use Filament\Actions\EditAction;
8 use Filament\Tables\Table;
9 use Filament\Tables\Columns\TextColumn;
10
11 class TagsTable
12 {
13 public static function configure(Table $table): Table
14 {
15 return $table
16 ->columns([
17 TextColumn::make('name')
18])
19 ->filters([
20 //
21])
22 ->recordActions([
23 EditAction::make(),
24])
25 ->toolbarActions([
26 BulkActionGroup::make([
27 DeleteBulkAction::make(),
28]),
29]);
30 }
31 }
32
```

Setelah itu admin dapat:

- Menambah tag
- Mengedit tag
- Menghapus tag



Tambahkan kode pada CreateTag.php dan EditTag.php pada folder pages agar setelah create atau edit selesai maka bisa langsung ke halaman index

PraktikumPWL > app > Filament > Admin > Resources > Tags > Pages > CreateTag.php

```
1 <?php
2
3 namespace App\Filament\Admin\Resources\Tags\Pages;
4
5 use App\Filament\Admin\Resources\Tags\TagResource;
6 use Filament\Resources\Pages\CreateRecord;
7
8 class CreateTag extends CreateRecord
9 {
10 protected static string $resource = TagResource::class;
11
12 protected function getRedirectUrl(): string
13 {
14 return $this->getResource()->getUrl('index');
15 }
16 }
17
```

PraktikumPWL > app > Filament > Admin > Resources > Tags > Pages > EditTag.php

```
1 <?php
2
3 namespace App\Filament\Admin\Resources\Tags\Pages;
4
5 use App\Filament\Admin\Resources\Tags\TagResource;
6 use Filament\Actions\DeleteAction;
7 use Filament\Resources\Pages>EditRecord;
8
9 class EditTag extends EditRecord
10 {
11 protected static string $resource = TagResource::class;
12
13 protected function getHeaderActions(): array
14 {
15 return [
16 DeleteAction::make(),
17];
18 }
19
20 protected function getRedirectUrl(): string
21 {
22 return $this->getResource()->getUrl('index');
23 }
24 }
```

## I. Menambahkan Relationship pada Model Post

Buka file post.php pada model dan hapus tags pada fillable

PraktikumPWL > app > Models > Post.php

```
1 <?php
2
3 namespace App\Models;
4
5 use Illuminate\Database\Eloquent\Model;
6
7 class Post extends Model
8 {
9 //
10 protected $fillable = [
11 'title',
12 'slug',
13 'category_id',
14 'color',
15 'image',
16 'body',
17 'tags',
18 'published',
19 'published_at',
20];
21 protected $casts = [
22 'tags' => 'array',
23 'published' => 'boolean',
24 'published_at' => 'date',
25];
26
27 public function category()
28 {
29 return $this->belongsTo(Category::class);
30 }
31 }
32
33
34
```

Tambahkan function pada model post

- public function tags()  
{  
    return \$this->belongsToMany(Tag::class,'post\_tag');  
}

```
PraktikumPWL > app > Models > Post.php
1 <?php
2
3 namespace App\Models;
4
5 use Illuminate\Database\Eloquent\Model;
6
7 class Post extends Model
8 {
9 //
10 protected $fillable = [
11 'title',
12 'slug',
13 'category_id',
14 'color',
15 'image',
16 'body',
17 'published',
18 'published_at',
19];
20 protected $casts = [
21 'tags' => 'array',
22 'published' => 'boolean',
23 'published_at' => 'date',
24];
25
26 public function category()
27 {
28 return $this->belongsTo(Category::class);
29 }
30
31 public function tags()
32 {
33 return $this->belongsToMany(Tag::class,'post_tag');
34 }
35
36 }
```

---

## 📌 J. Menambahkan Relationship pada Model Tags

Tambahkan fungsi pada **Model Tag**

- public function posts()  
{

```

return $this->belongsToMany(Post::class,'post_tag');
}

```

```

PraktikumPWL > app > Models > Tag.php
1 <?php
2
3 namespace App\Models;
4
5 use Illuminate\Database\Eloquent\Model;
6
7 class Tag extends Model
8 {
9 protected $fillable = [
10 'name'
11];
12
13 public function posts()
14 {
15 return $this->belongsToMany(Post::class,'post_tag');
16 }
17
18 }
19

```

## K. Menggunakan Relationship pada Form Post

Buka **PostResource** → **Form Schema**

Ganti field tags dengan:

- `Select::make('tags')`  
     `->relationship('tags','name')`  
     `->multiple()`

```

PraktikumPWL > app > Filament > Admin > Resources > Posts > Schemas > PostForm.php
22 {
23 {
24 return $schema
25 ->components([
26 //section 3 - meta
27 Section::make("Meta Information")
28 ->schema([
29 // TagsInput::make("tags"),
30 Select::make("tags")
31 ->relationship("tags","name"),
32 Checkbox::make("published"),
33 DateTimePicker::make("published_at"),
34]),
35]->columnSpan(1)
36]->columns(3);
37 }
38 }
39

```

Penjelasan:

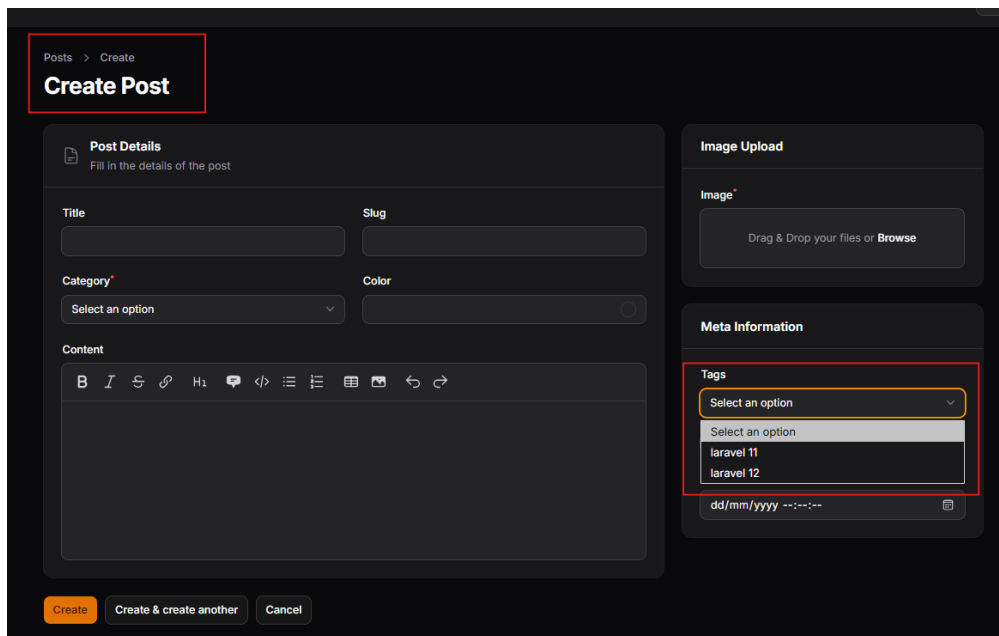
| Method       | Fungsi                |
|--------------|-----------------------|
| relationship | mengambil data relasi |
| multiple     | memilih banyak data   |

## 📌 L. Hasil Form Post

Dropdown tag akan menampilkan:

- Laravel 10
- Laravel 11
- Laravel 12

User dapat memilih lebih dari satu tag.



Tambahkan function `multiple()` untuk membuat **field Select bisa memilih lebih dari satu data**.

PraktikumPWL > app > Filament > Admin > Resources > Posts > Schemas > PostForm.php

```
22 {
23 {
24 return $schema
25 ->components([
26 //section 3 - meta
27 Section::make("Meta Information")
28 ->schema([
29 // TagsInput::make("tags"),
30 Select::make("tags")
31 ->relationship("tags", "name")
32 ->multiple()
33 ->preload(),
34 Checkbox::make("published"),
35 DateTimePicker::make("published_at"),
36]),
37]->columnSpan(1)
38]->columns(3);
39 }
40 }
```

Posts > Create

## Create Post

**Post Details**

Fill in the details of the post

Title

Slug

Category\*

Color

Content

B

I

U

Link

H1

Code

List

Table

Image

Undo

Redo

Create

Create & create another

Cancel

**Image Upload**

Image\*

Drag & Drop your files or **Browse**

**Meta Information**

Tags

laravel 11 x laravel 12 x

☐ Published

Published at

dd/mm/yyyy --:--:--

## 📌 M. Membuat Relationship Manager

Relationship Manager memungkinkan mengelola tag langsung dari post.

Jalankan perintah:

- `php artisan make:filament-relation-manager`

Isi parameter:

Resource : PostResource

Relationship : tags

Column : name

```
PS C:\laragon\www\PraktikumPwL> php artisan make:filament-relation-manager PostResource tags name

Filament can link this to an existing resource, which will open the resource's pages instead of modals when links are clicked. It will also inherit the resource's configuration.

Do you want to do this? (yes/no) [no]
> no

Would you like to use an existing form schema class? (yes/no) [no]
> no

Should the configuration be generated from the current database columns? (yes/no) [no]
> no

Should there be a read-only "view" modal on the relation manager? (yes/no) [no]
> no

Would you like to use an existing table class? (yes/no) [no]
> no

Does the model use soft-deletes? (yes/no) [no]
> no

What type of relationship is this?
HasMany Illuminate\Database\Eloquent\Relations\HasMany
BelongsToMany Illuminate\Database\Eloquent\Relations\BelongsToMany
MorphMany Illuminate\Database\Eloquent\Relations\MorphMany
MorphToMany Illuminate\Database\Eloquent\Relations\MorphToMany
Other other
> BelongsToMany
BelongsToMany

[INFO] Filament relation manager [App\Filament\Admin\Resources\Posts\RelationManagers\TagsRelationManager] created successfully.

[INFO] Make sure to register the relation in [App\Filament\Admin\Resources\Posts\PostResource::getRelations()].
```

## ✦ N. Menghubungkan Relationship Manager

Buka:

- PostResource.php

Tambahkan:

- public static function getRelations(): array  
{  
    return [  
        TagsRelationManager::class,  
    ];  
}

PraktikumPWL > app > Filament > Admin > Resources > Posts > PostResource.php

```
14 use Filament\Tables\Table;
15 use BackedEnum;
16 use App\Filament\Admin\Resources\Posts\RelationManagers\TagsRelationManager;
17
```

```
Windsurf: Refactor | Explain
18 class PostResource extends Resource
19 {
20 protected static ?string $model = Post::class;
21
22 Windsurf: Refactor | Explain | Generate Function Comment | X
23 public static function getNavigationIcon(): string|null
24 {
25 return 'heroicon-s-briefcase';
26
27 Windsurf: Refactor | Explain | Generate Function Comment | X
28 public static function form(Schema $schema): Schema
29 {
30
31 Windsurf: Refactor | Explain | Generate Function Comment | X
32 public static function table(Table $table): Table
33 {
34 return PostsTable::configure($table);
35
36 Windsurf: Refactor | Explain | Generate Function Comment | X
37 public static function getRelations(): array
38 {
39 return [
40 //
41 TagsRelationManager::class,
42],
43 }
44
45 Windsurf: Refactor | Explain | Generate Function Comment | X
46 public static function getPages(): array
47 {
48 return [
49 'index' => ListPosts::route('/'),
50 'create' => CreatePost::route('/create'),
51 'edit' => EditPost::route("/{record}/edit"),
52];
53 }
54 }
```



## O. Fitur Relationship Manager

Buka file TagsRelationManager.php dan modifikasi codenya

PraktikumPWL > app > Filament > Admin > Resources > Posts > RelationManagers > TagsRelationManager.php

```
20 {
24 {
25 return $schema;
26 }
27 }
28
29 Windsurf: Refactor | Explain | Generate Function Comment | X
30 public function table(Table $table): Table
31 {
32 return $table
33 ->recordTitleAttribute('name')
34 ->columns([
35 TextColumn::make('id'),
36 TextColumn::make('name')
37 // ->searchable(),
38])
39 ->filters([
40 //
41])
42 ->headerActions([
43 CreateAction::make(),
44 AttachAction::make(),
45])
46 ->recordActions([
47 EditAction::make(),
48 DetachAction::make(),
49 DeleteAction::make(),
50])
51 ->toolbarActions([
52 // BulkActionGroup::make([
53 // DetachBulkAction::make(),
54 // DeleteBulkAction::make(),
55 //]),
56]);
57 }
58 }
```

Posts > Edit

### Edit Post

Post Details

Fill in the details of the post

Title: laravel Slug: crud

Category: laravel Color: #d44343

Content

Image Upload

Image: 09K3K2M0QRX1B8PYJXNQYBPW3.jpg

Meta Information

Tags: laravel 11 laravel 12

Published: ☒ Published

Published at: 07/03/2026 00:00:00

Save changes Cancel

Tags

Now Tag Attach

| Id | Name       |                                                                    |
|----|------------|--------------------------------------------------------------------|
| 1  | laravel 11 | <a href="#">Edit</a> <a href="#">Detach</a> <a href="#">Delete</a> |
| 2  | laravel 12 | <a href="#">Edit</a> <a href="#">Detach</a> <a href="#">Delete</a> |

Showing 1 to 2 of 2 results

Per page: 10

## P. Contoh Many-to-Many Relationship

Messages > Create

### Create Message

Message text

Some important message.

CreateCreate & create anotherCancel

Contacts

Garrett Douglas × Loma Kuhic ×

Select an option

Graciela Aufderhar  
Prof. Jesse Sawayn Sr.  
Suzanne Orn

Dashboard

Categories

Products

Tags

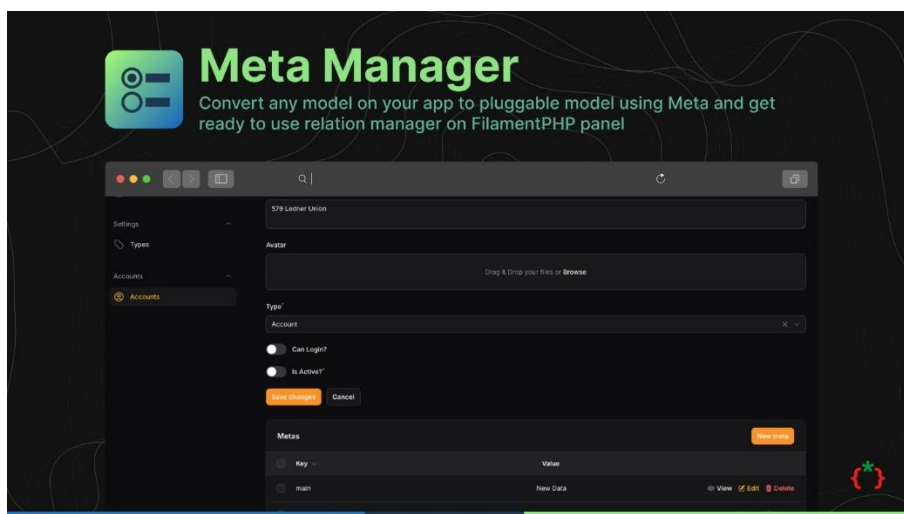
## Tags

New tag

| <input type="checkbox"/> | Name        |                                             |
|--------------------------|-------------|---------------------------------------------|
| <input type="checkbox"/> | on sale     | <a href="#">Edit</a> <a href="#">Delete</a> |
| <input type="checkbox"/> | new edition | <a href="#">Edit</a> <a href="#">Delete</a> |
| <input type="checkbox"/> | trending    | <a href="#">Edit</a> <a href="#">Delete</a> |

Showing 1 to 3 of 3 results

Per page10



## ✦ Q. Contoh Attach dan Detach Tag

### Attach Tag

Attach → Laravel 11

### Detach Tag

Detach → Laravel 12

Pivot table otomatis berubah:

post\_tag

-----

post\_id

tag\_id

---

## ✦ R. Keuntungan Many-to-Many Relationship

| JSON Column        | Many-to-Many            |
|--------------------|-------------------------|
| Tidak terstruktur  | Struktur database jelas |
| Data duplikat      | Normalisasi database    |
| Query sulit        | Query mudah             |
| Sulit dimodifikasi | Mudah dikelola          |

---

## ✦ S. Latihan Praktikum

1. Buat tabel **tags**
2. Buat pivot table **post\_tag**
3. Buat model **Tag**
4. Tambahkan relationship pada model Post
5. Buat **Tag Resource**
6. Tambahkan **multiple select tags** pada Post form
7. Implementasikan **Tags Relationship Manager**

---

### Screenshot yang dikumpulkan

1. Halaman **Tag Resource**
2. Form Post dengan multiple tags
3. Relationship Manager Tags

#### 4. Pivot table **post\_tag**

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#### **T. Analisis & Diskusi**

1. Apa perbedaan **HasMany** dan **Many-to-Many**?
  2. Mengapa pivot table diperlukan?
  3. Apa fungsi attach dan detach pada Filament?
  4. Mengapa JSON column kurang baik untuk relasi?
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#### **U. Kesimpulan**

Pada praktikum ini mahasiswa telah mempelajari:

- Konsep **Many-to-Many Relationship**
- Implementasi pivot table pada Laravel
- Relationship model menggunakan belongsToMany
- Multiple select relationship pada Filament
- Penggunaan Relationship Manager untuk mengelola relasi